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Inter-VLAN Routing Using a Layer 3 (Multilayer) Switch

VLAN Creation and Routing via Switch Virtual Interfaces (SVIs)

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Platform: Cisco Packet Tracer

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1. Overview

This report documents a project on inter-VLAN routing implemented entirely on a single Layer 3 (multilayer) switch, using Switch Virtual Interfaces (SVIs) rather than an external router. This is a third method of inter-VLAN routing, alongside the two-physical-interface method and Router-on-a-Stick documented in earlier phases of the McNasir Systems network project, and represents the approach generally preferred in enterprise environments where the hardware is available.

2. Advantages and Disadvantages

A Layer 3 switch performs routing in hardware using SVIs, eliminating the need for a separate router and trunk link entirely:

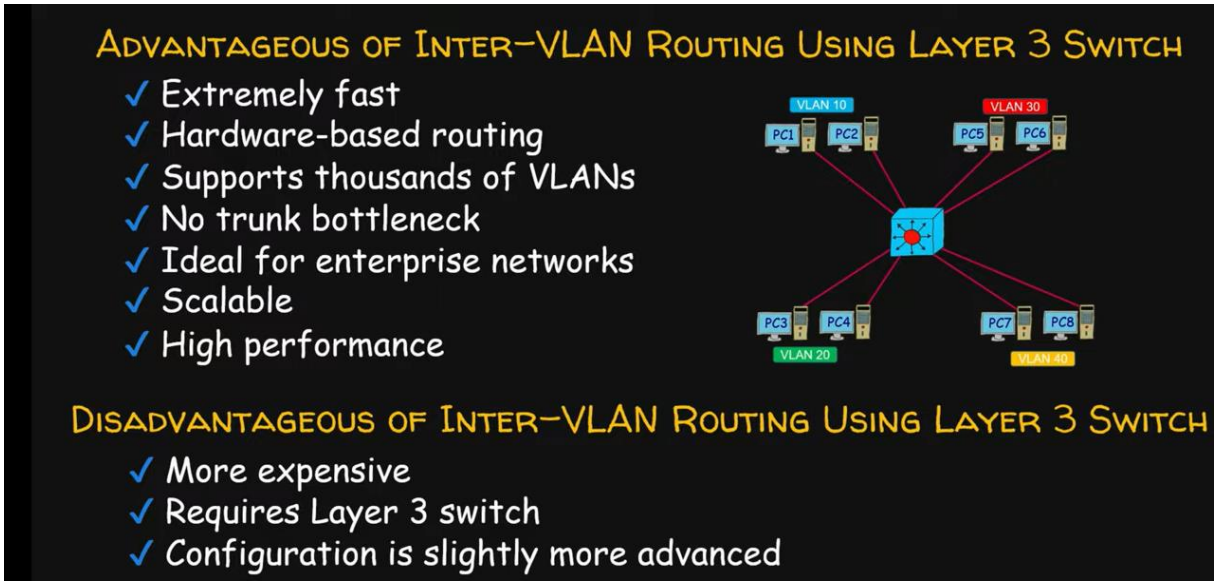


Figure 1 — Advantages and Disadvantages of Inter-VLAN Routing Using a Layer 3 Switch

3. Implementation Steps

- Step 1: Create VLANs on the switch.
- Step 2: Assign switch interfaces to their respective VLANs.
- Step 3: Create Switch Virtual Interfaces (SVIs) — one per VLAN.
- Step 4: Enable routing on the switch (ip routing).
- Step 5: Configure PC IP addresses and default gateways.
- Step 6: Verify VLAN membership, SVI status, routing table, and end-to-end connectivity.

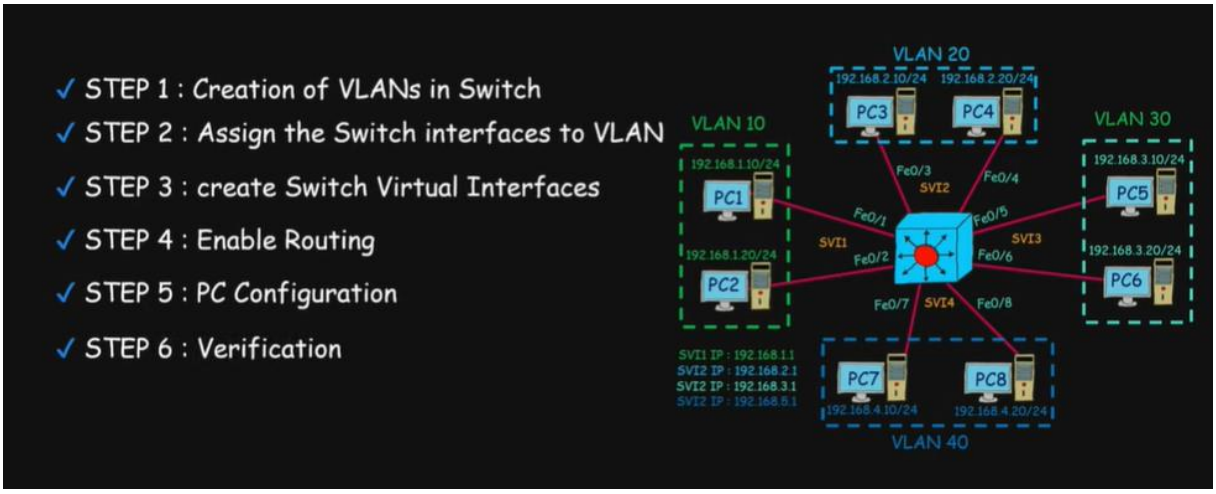


Figure 2 — Implementation Steps and Design Reference

4. Network Design

4.1 Addressing Scheme

- VLAN 10 (HR): 192.168.1.0/24 — SVI Gateway 192.168.1.1
- VLAN 20 (IT): 192.168.2.0/24 — SVI Gateway 192.168.2.1
- VLAN 30 (Sales): 192.168.3.0/24 — SVI Gateway 192.168.3.1
- VLAN 40 (Finance): 192.168.4.0/24 — SVI Gateway 192.168.4.1

4.2 Devices

- Multilayer Switch1 — performs both switching and routing; no external router required
- PC1, PC2 — VLAN 10 (HR)
- PC3, PC4 — VLAN 20 (IT)
- PC5, PC6 — VLAN 30 (Sales)
- PC7, PC8 — VLAN 40 (Finance)

5. Network Diagram

The diagram below shows the completed topology, with each department's VLAN, IP addressing, and its corresponding SVI gateway.

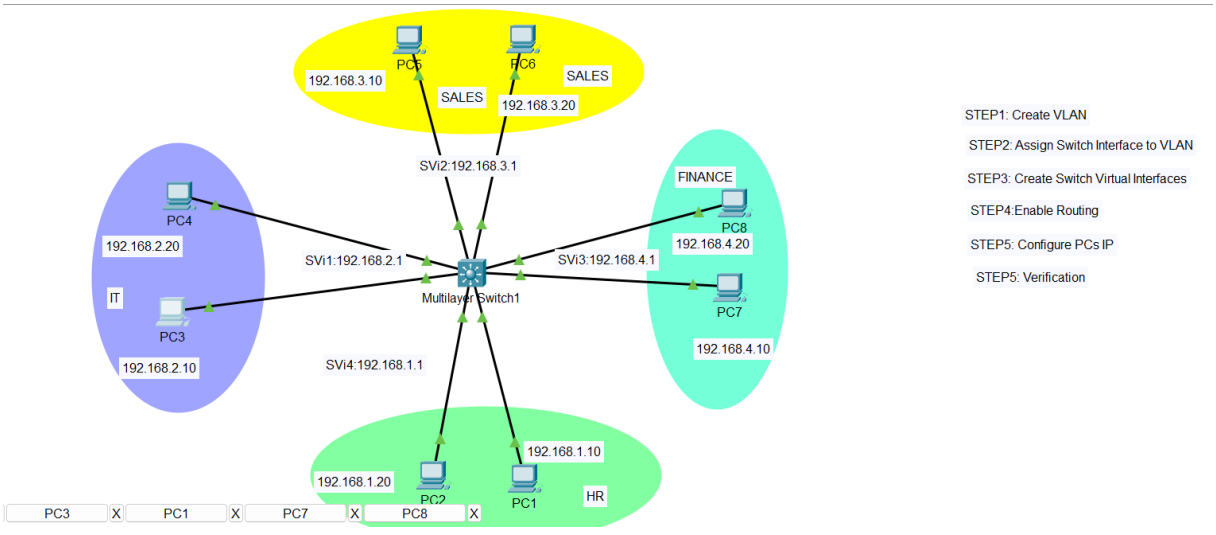


Figure 3 — Packet Tracer Network Topology

6. Switch Verification

6.1 SVI Status and Routing Table

show ip interface brief confirms all four SVIs (Vlan10–Vlan40) are up/up with their correct IP addresses, and ip routing has been enabled. The subsequent show ip route output confirms all four subnets appear as directly connected (C), meaning the switch is aware of and able to route between all four VLANs.

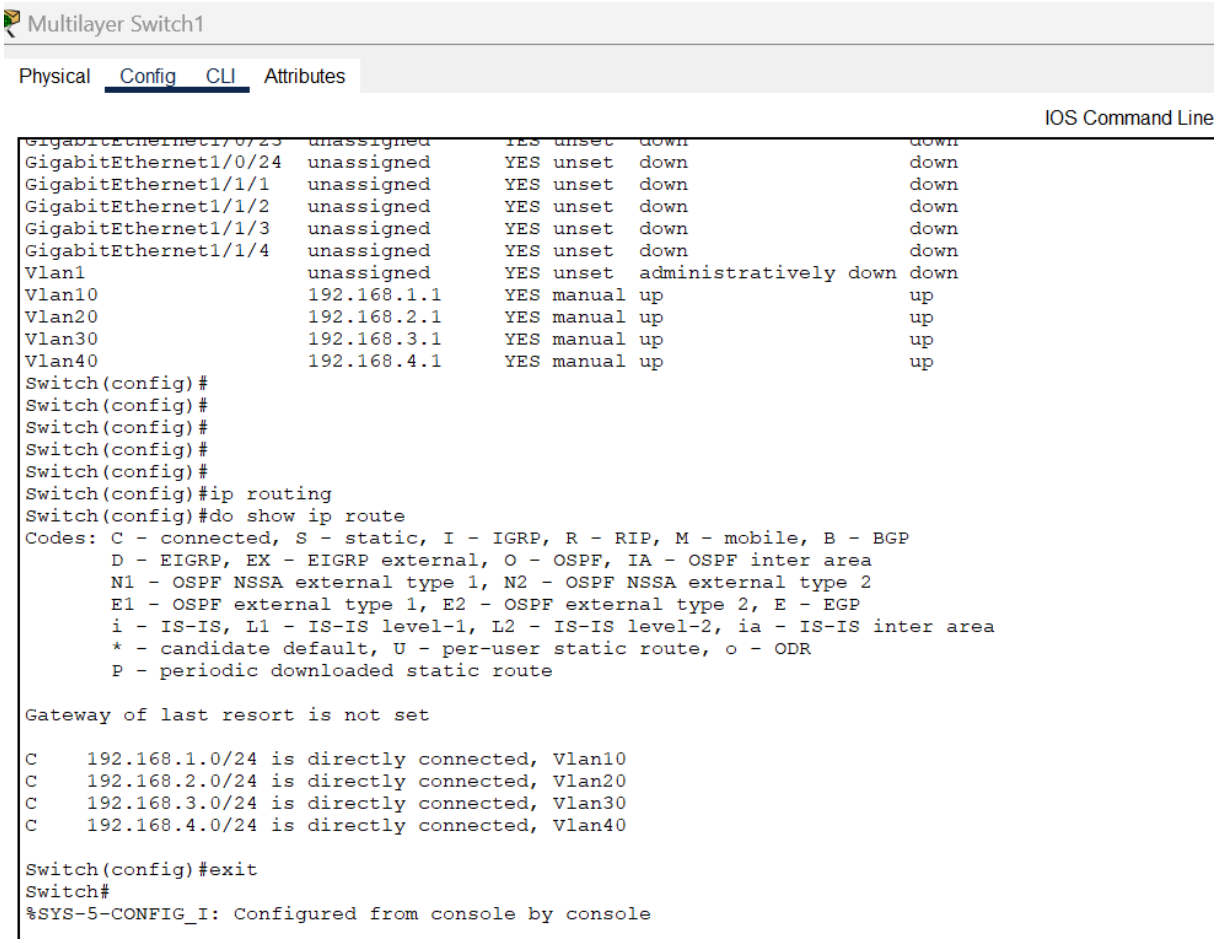


Figure 4 — show ip interface brief and show ip route (Multilayer Switch1)

6.2 VLAN Membership

show vlan brief confirms all four VLANs are active with the correct ports assigned: VLAN 10 (HR) on Gig1/0/1–2, VLAN 20 (IT) on Gig1/0/3–4, VLAN 30 (Sales) on Gig1/0/5–6, and VLAN 40 (Finance) on Gig1/0/7–8.

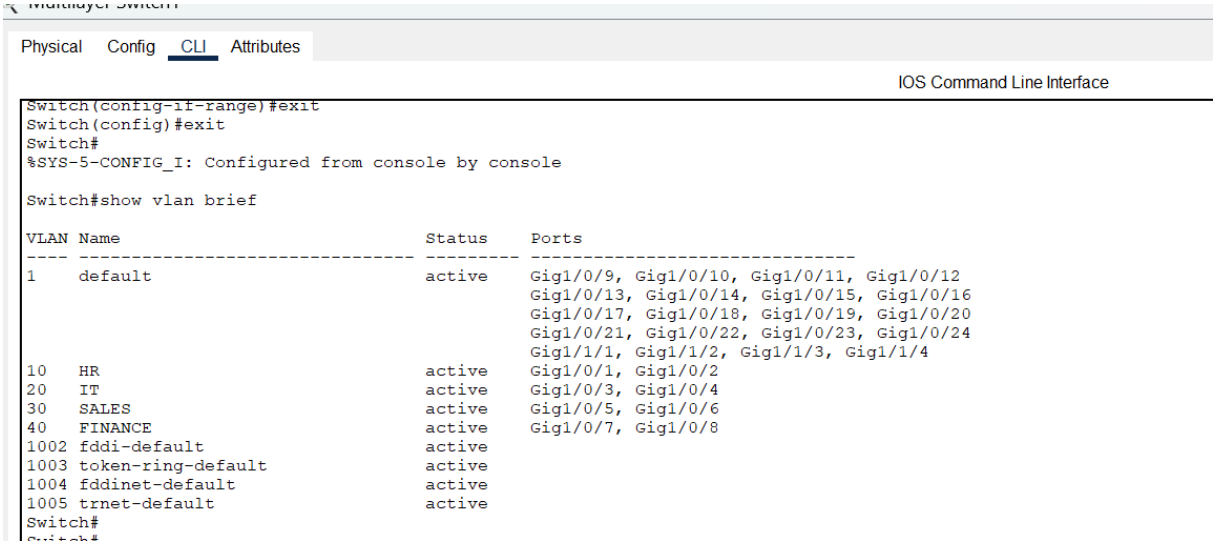


Figure 5 — show vlan brief (Multilayer Switch1)

7. Connectivity Verification

7.1 Within-VLAN Connectivity

VLAN 10 (HR) — Ping from PC1 to PC2, successful through the switch's Layer 2 forwarding:

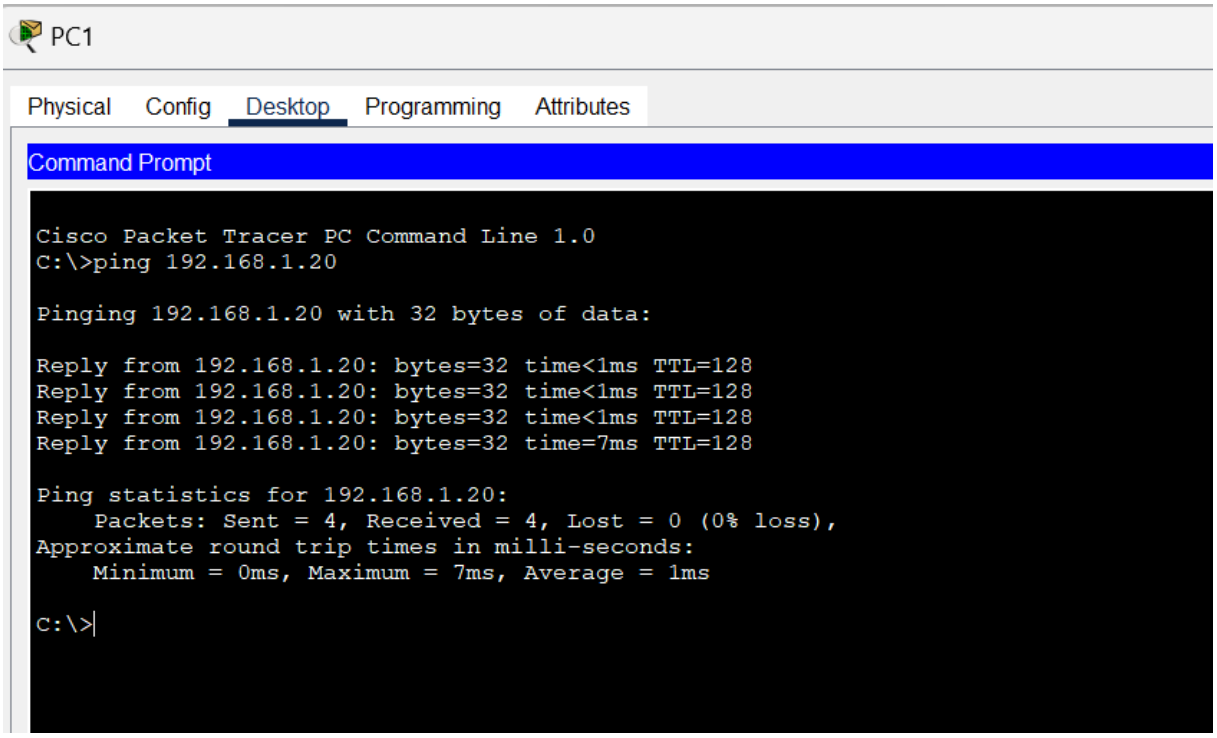


Figure 6 — Ping PC1 to PC2 (VLAN 10 / HR)

VLAN 20 (IT) — Ping from PC3 to PC4, successful through the switch's Layer 2 forwarding:

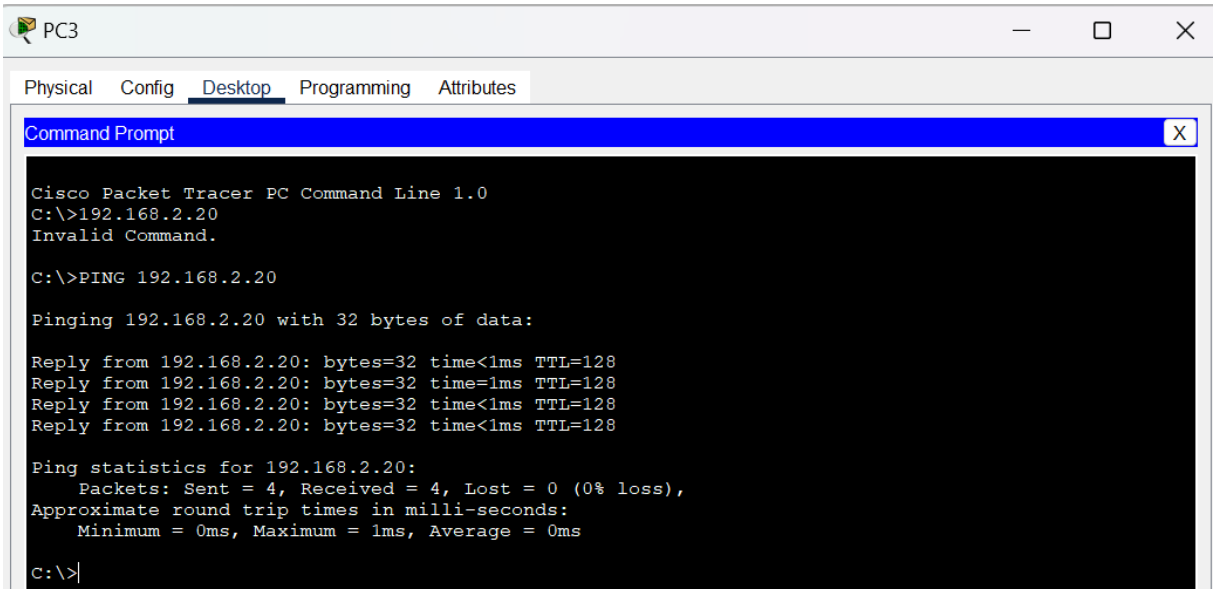


Figure 7 — Ping PC3 to PC4 (VLAN 20 / IT)

VLAN 30 (Sales) — Ping from PC5 to PC6, successful through the switch's Layer 2 forwarding:

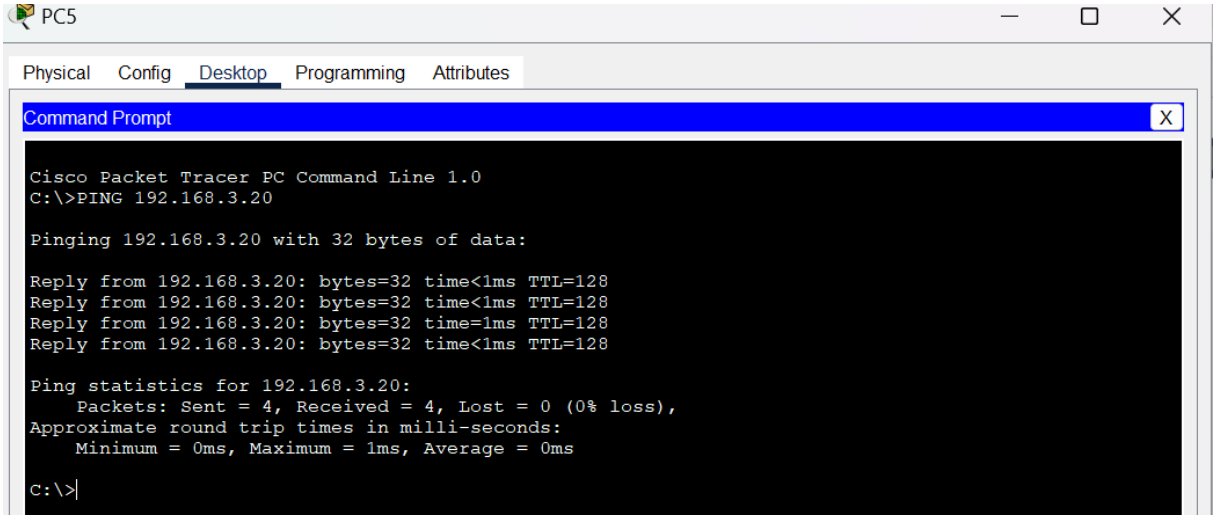


Figure 8 — Ping PC5 to PC6 (VLAN 30 / Sales)

VLAN 40 (Finance) — Ping from PC7 to PC8, successful through the switch's Layer 2 forwarding:

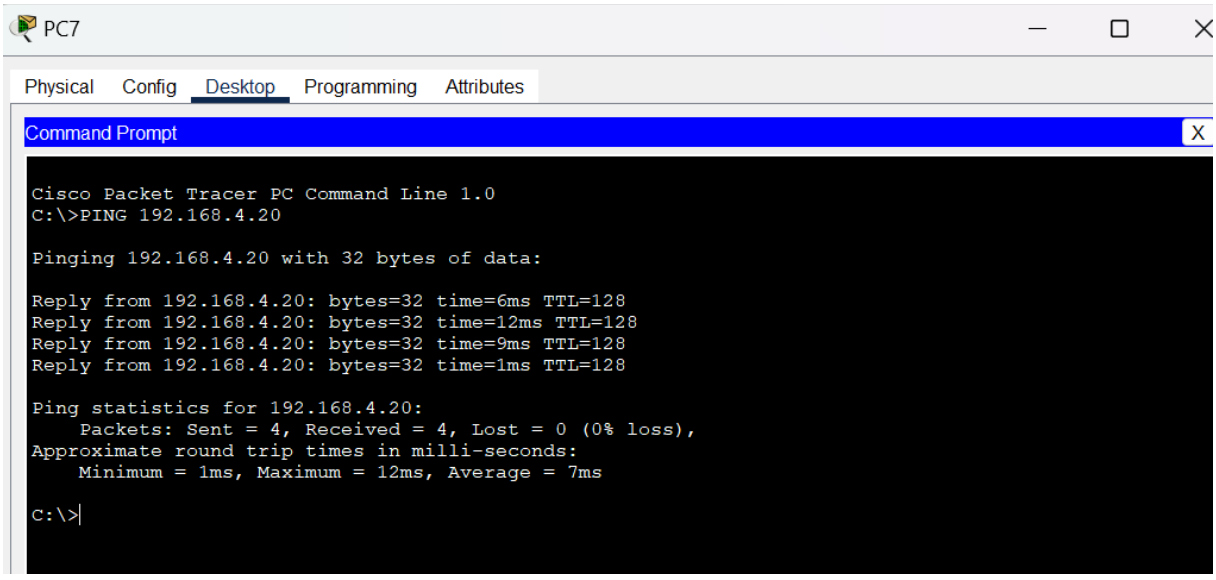


Figure 9 — Ping PC7 to PC8 (VLAN 40 / Finance)

7.2 Inter-VLAN Connectivity

VLAN 40 (Finance) to VLAN 10 (HR) — Ping from PC8 to PC1, routed through the switch's Layer 3 SVI. The first attempt shows request timeouts, which is expected: it reflects the ARP resolution delay for the very first packet between two devices that have not yet communicated. The second attempt confirms successful routing between VLANs:

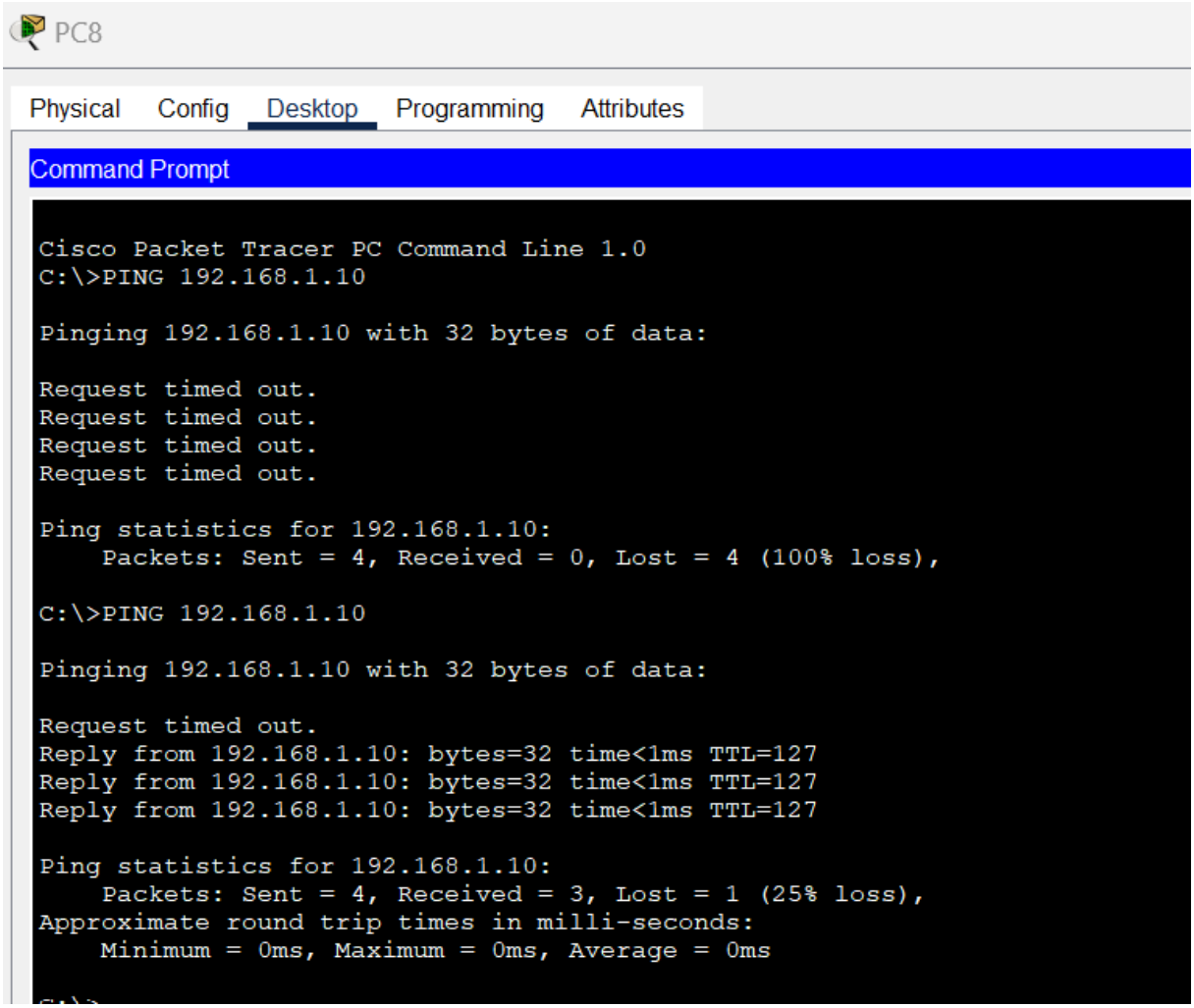


Figure 10 — Ping PC8 to PC1 (VLAN 40 → VLAN 10, via Layer 3 routing)

VLAN 20 (IT) to VLAN 30 (Sales) — Ping from PC3 to PC5 and PC6, routed through the switch's Layer 3 SVI, successful (the single lost packet on the first destination reflects the same initial ARP delay):

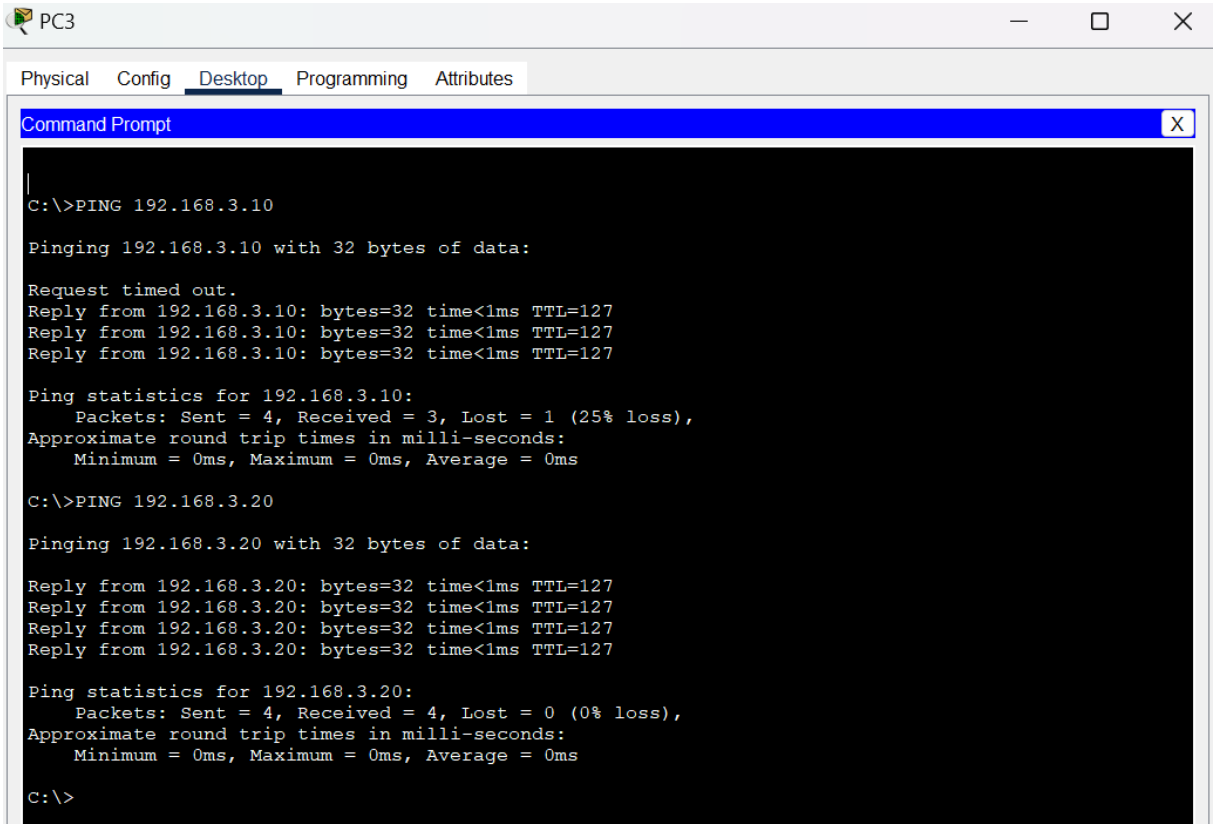


Figure 11 — Ping PC3 to PC5/PC6 (VLAN 20 → VLAN 30, via Layer 3 routing)

8. Outcome

All four VLANs — HR, IT, Sales, and Finance — were successfully created and routed entirely on a single multilayer switch using SVIs, with no external router required. Within-VLAN connectivity was confirmed for all four departments, and inter-VLAN connectivity was confirmed in both directions tested. This validates the Layer 3 switch as a working, scalable alternative to both the two-interface method and Router-on-a-Stick for McNasir Systems' inter-VLAN routing needs.

9. Conclusion

Of the three inter-VLAN routing methods explored across this project, the Layer 3 switch (SVI) approach offers the strongest combination of performance and scalability: routing happens in hardware at wire speed, no trunk link or external router is required, and additional VLANs can be added simply by creating a new SVI. The primary trade-off is cost, since multilayer switches carry a higher price point than standard Layer 2 switches — a factor to weigh against McNasir Systems' expanding budget and long-term growth plans.